Trust Signal Health Pipeline:

A Structured Governance Diagnostic for Detecting Proxy-Outcome Divergence in AI-Optimized Operations

Gigi Aulabaugh

PixelKraze LLC

NovaWireless Synthetic Research Lab • March 2026

Abstract

This paper documents the Trust Signal Health Pipeline, a seven-layer governance diagnostic that scores every call in the NovaWireless synthetic call center on five trust signal dimensions, aggregates signals to representative, scenario, and customer levels, tracks month-over-month drift, evaluates configurable WATCH/VETO thresholds, and validates whether trust scores predict customer churn. Applied to 82,165 call records across 12 months, the pipeline reveals a 42.8 percentage point Goodhart gap (89.3% proxy resolution vs. 46.5% true resolution), a bimodal trust score distribution (mean 70.8, median 76.3), and stable drift velocity confirming a steady-state lie rather than progressive degradation. The churn integration analysis yields a trust-churn correlation of $r = -0.328$ at the decile level, indicating that while the trust score captures structural divergence, churn signal remains diffuse at the call level—consistent with the broader research finding that proxy metrics are informationally neutral for predicting customer retention. The pipeline produces 9 diagnostic visualizations, 5 structured reports, and a machine-readable governance JSON suitable for downstream consumption by AI optimization systems.

Keywords: governance pipeline, trust scoring, proxy metrics, Goodhart's Law, KPI integrity, AI governance, call center operations, churn prediction

Introduction

When a contact center optimizes for First Contact Resolution (FCR), representatives learn to game the proxy flag—marking calls resolved without fixing the underlying problem. The dashboard stays green while customer outcomes deteriorate. This is Goodhart's Law (Goodhart, 1975) applied to operational AI: when a measure becomes a target, it ceases to be a good measure. The companion paper *When KPIs Lie* (Aulabaugh, 2026a) demonstrates that pressure amplifies this gaming without moving the proxy metric, and the *Goodhart's Law Addendum* (Aulabaugh, 2026b) documents a 20.2pp proxy-truth gap that remains stable across 12 months.

This paper documents the Trust Signal Health Pipeline—the structured governance diagnostic that detects and quantifies this divergence through call metadata rather than transcript language. Where the companion *novawireless-transcript-analysis* repository detects divergence through NLP features (profanity rates, scenario-distinctive terms, target-specific lift profiles), this pipeline detects it through resolution flags, credit patterns, detection signals, and representative behavioral profiles. Together, the two signal families provide independent confirmation that the system is lying.

The pipeline was designed to be composable: it produces a machine-readable governance JSON that downstream systems (including AI optimization loops) can consume to decide whether their training labels are trustworthy. If the pipeline says VETO, the labels should not be used for optimization. This is the core governance function.

Pipeline Architecture

The pipeline processes 12 monthly *calls_sanitized_YYYY-MM.csv* files through seven sequential layers: integrity gate, trust signal scoring, multi-level aggregation, monthly trend tracking, threshold alerts, customer churn integration, and evidence artifact generation. Each layer writes structured output to organized subdirectories: *output/data/* for CSVs,

output/figures/ for diagnostic charts, and *output/reports/* for JSON and text reports.

Layer 1: Integrity Gate

The integrity gate validates every row against structural expectations: required columns present, scenario labels in the known set, credit types valid, detection flag consistency, and duplicate call_id handling. Hard failures (missing data, unknown scenarios) are quarantined but preserved for audit. Soft analytical signals—specifically proxy-true divergence and detection flags on clean scenarios—are flagged but not quarantined, because they represent the Goodhart signal itself rather than data quality problems. In the current dataset, all 82,165 rows pass the hard gate (0% quarantine rate), while 44.0% carry the proxy-true divergence soft flag.

Layer 2: Trust Signal Scoring

Each call receives five component scores that sum to a composite trust score on a 0–100 scale. The proxy-versus-true gap (25 points) fires when the proxy flag says resolved but ground truth says not—the direct Goodhart signal. Detection signal density (20 points) measures the fraction of six fraud/gaming indicators that fire per call. Credit risk score (20 points) captures unauthorized credit patterns, with bandaid credits receiving the maximum 1.0 risk weight. Rep drift score (15 points) combines gaming propensity (65% weight) and burnout (35% weight) into a behavioral risk composite. Outcome risk score (20 points) aggregates downstream consequences: unresolved calls, repeat contacts, and escalations.

Layers 3–7: Aggregation Through Evidence

Layer 3 aggregates call-level scores to representative (250 agents), scenario (10 categories), and customer (5,000 accounts) summaries. Layer 4 tracks all governance signals month-over-month and computes drift velocity—the rate of change per month—to distinguish progressive degradation from steady-state divergence. Layer 5 evaluates configurable

WATCH/VETO thresholds at scenario and representative levels, plus drift velocity alerts for months where trust drops faster than 2 points. Layer 6 validates whether the trust score predicts customer churn by computing churn rates across trust score deciles. Layer 7 generates nine diagnostic charts, a narrative summary report, a structured governance JSON, and threshold alert reports.

Results

Trust Score Distribution

The composite trust score distribution across 82,165 calls reveals a bimodal pattern (Figure 1). The mean score is 70.8 and the median is 76.3—the rightward skew of the median reflects a large cluster of clean calls scoring above 80, which pulls the median away from the mean. The left tail (scores below 60) is populated almost entirely by gaming, fraud, and complex unresolvable scenarios. The 10th percentile falls at 40.2, indicating that roughly one in ten calls carries severe trust compromise.

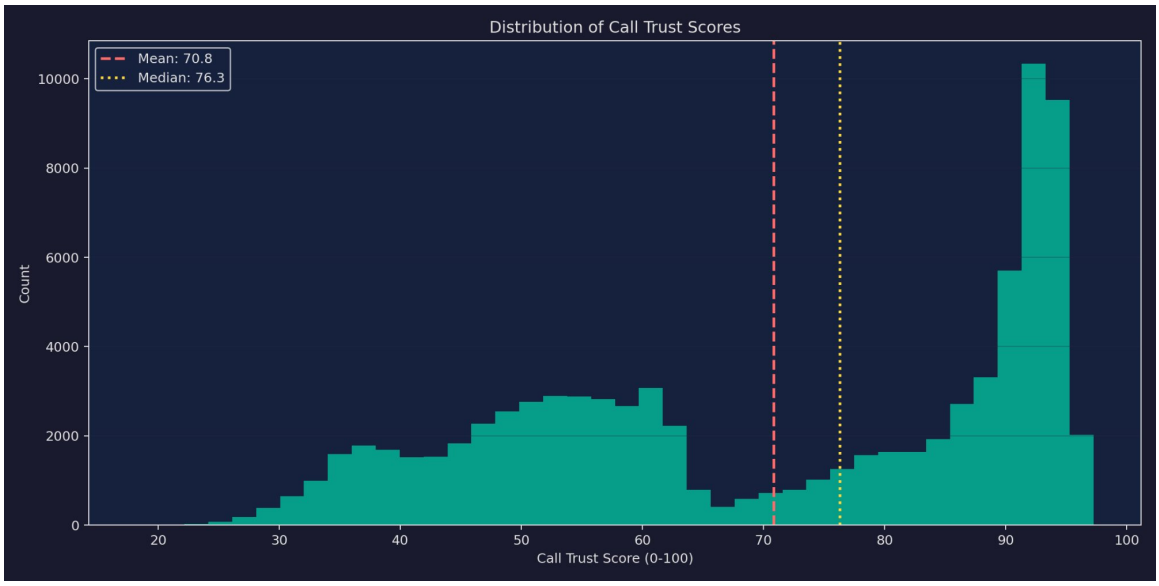

Figure 1. *Distribution of call trust scores (0–100). The bimodal pattern separates trustworthy clean calls (right cluster) from gaming/fraud calls (left tail). Mean = 70.8, median = 76.3.*

The Goodhart Gap by Scenario

Figure 2 shows the proxy versus true resolution rate for each of the 10 scenario types. The gamed_metric scenario exhibits the largest gap: 98.2% proxy resolution against 13.8% true resolution—an 84.3 percentage point divergence. All four fraud scenarios show similar patterns, with proxy rates above 90% and true rates below 25%. Even clean calls carry a 12.2pp gap, reflecting the baseline level of measurement noise in the proxy flag. The overall gap across all calls is 42.8pp (89.3% proxy vs. 46.5% true).

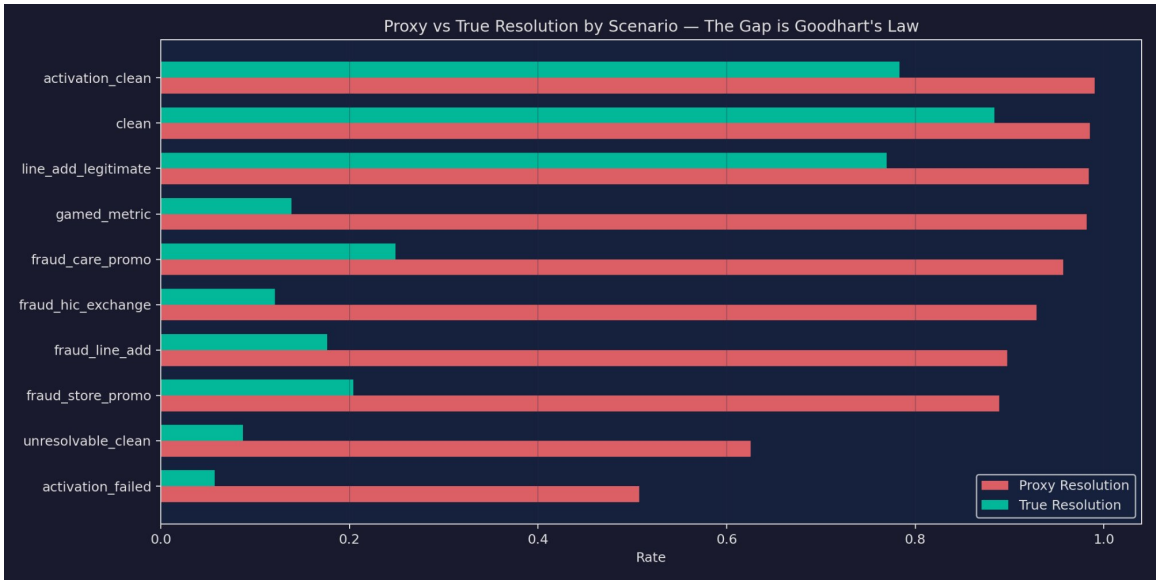

Figure 2. *Proxy versus true resolution rate by scenario. The gap between bars is Goodhart's Law made visible. Gaming and fraud scenarios show near-total proxy inflation.*

Scenario Signal Concentration

Figure 3 maps five governance signals across all 10 scenarios as a heatmap, revealing where drift concentrates. The gamed_metric scenario is the only one with elevated bandaid rate (0.71), confirming that bandaid credits are the mechanism of proxy gaming—unauthorized credits issued to suppress repeat contacts without resolving the customer's actual problem. Fraud scenarios show high detection density (0.30–0.61) and credit applied rates (0.85–0.96), while clean scenarios cluster near zero on all risk signals.

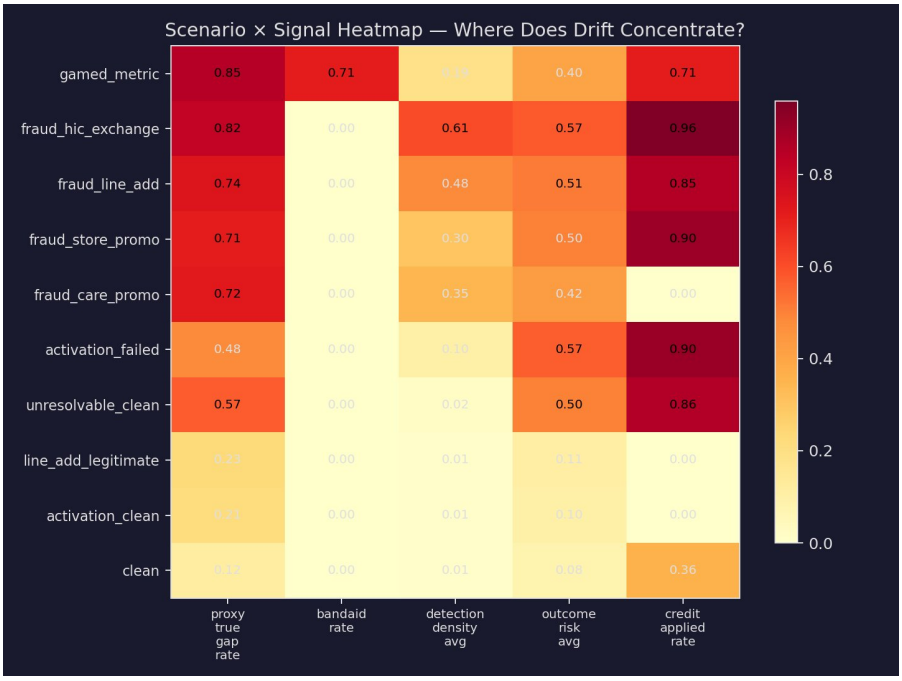

Figure 3. *Scenario × signal heatmap. Darker cells indicate higher risk. Gaming concentrates bandaid credits; fraud concentrates detection density and credit volume.*

Credit Type Analysis

Figure 4 decomposes the credit type mix by scenario for all credited calls. The gamed_metric scenario is dominated by bandaid credits (~71% of its credits), making the gaming mechanism structurally visible through metadata alone. Fraud scenarios show a mix of dispute credits and fee waivers consistent with unauthorized account manipulation. Clean and failed scenarios show proportional credit distributions without bandaid concentration.

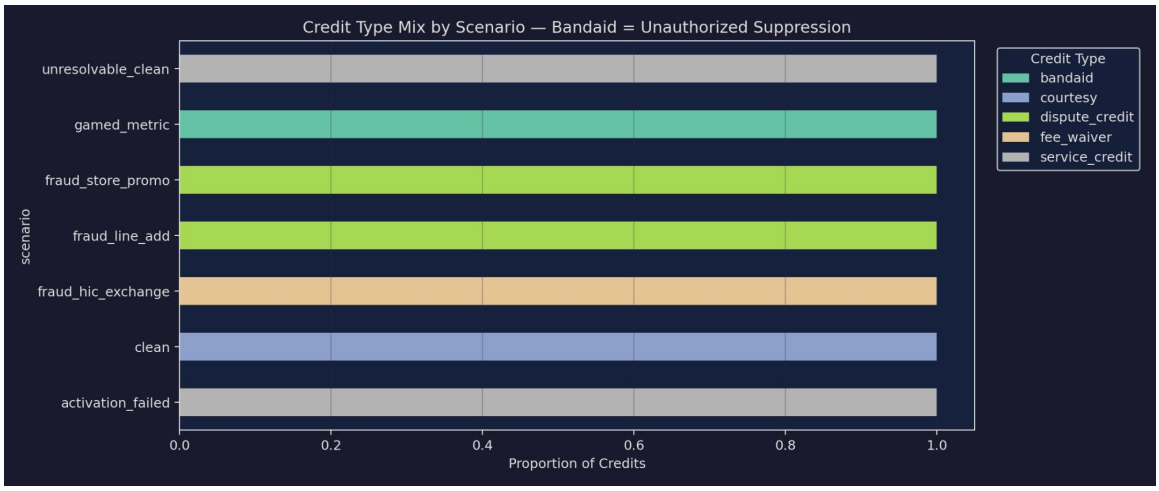

Figure 4. *Credit type mix by scenario. Bandaid credits (teal) concentrate exclusively in the gamed_metric scenario, providing a clear structural signature.*

Representative Trust Landscape

Figure 5 plots each of the 250 representatives as a point in trust score × call volume space, colored by resolution gap. The scatter confirms that trust score variation is independent of workload ($r = -0.06$ between calls handled and trust score), ruling out volume as a driver of poor outcomes. The color gradient shows that low-trust representatives (bottom of chart) consistently carry high resolution gaps (warm colors), while high-trust representatives show small gaps (green). The lowest-trust representative (REP00090, trust = 64.9) handles a normal volume of 329 calls but carries a 52.3% resolution gap.

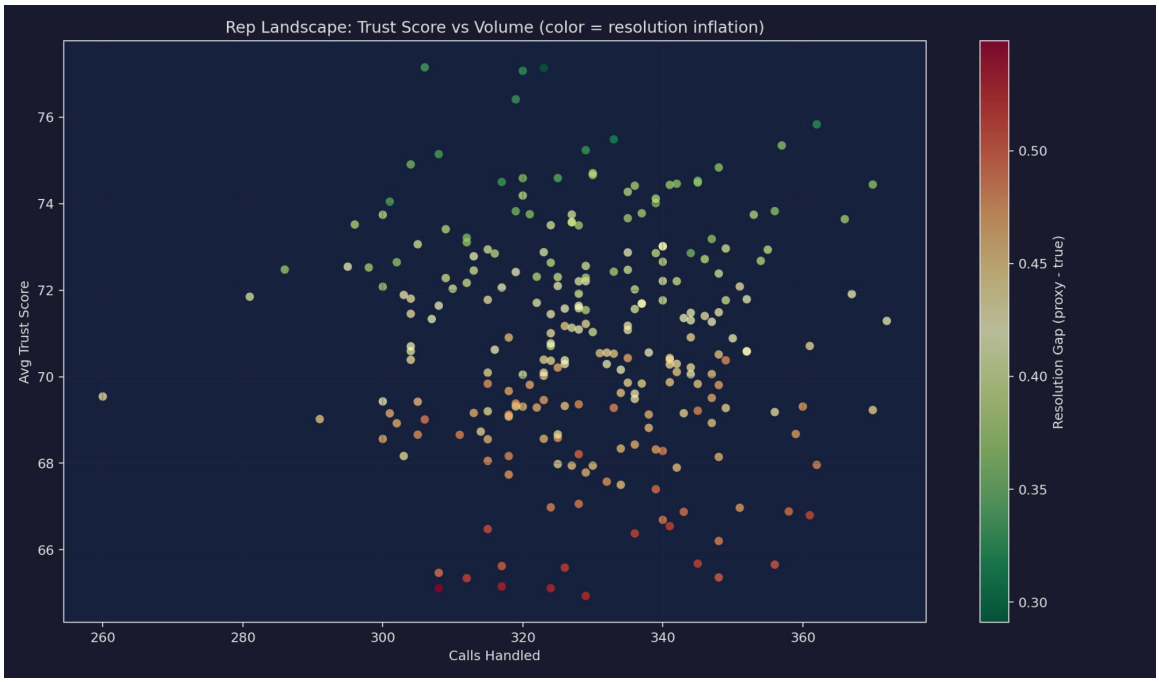

Figure 5. *Rep landscape: trust score vs. call volume, colored by resolution gap. Trust is driven by behavioral profile (gaming propensity, burnout), not workload.*

Rep-Level Signal Correlations

Figure 6 presents the full correlation matrix of rep-level signals. The key structural relationships are: gaming propensity correlates strongly with resolution gap ($r = 0.76$) and

inversely with trust score (r = –0.86); true resolution rate correlates inversely with proxy-true gap (r = –0.92) and positively with trust score (r = 0.88); and call volume shows near-zero correlation with every signal (all $|r| < 0.10$). The rep drift score (gaming + burnout composite) is the strongest single predictor of trust score at the rep level (r = –0.89).

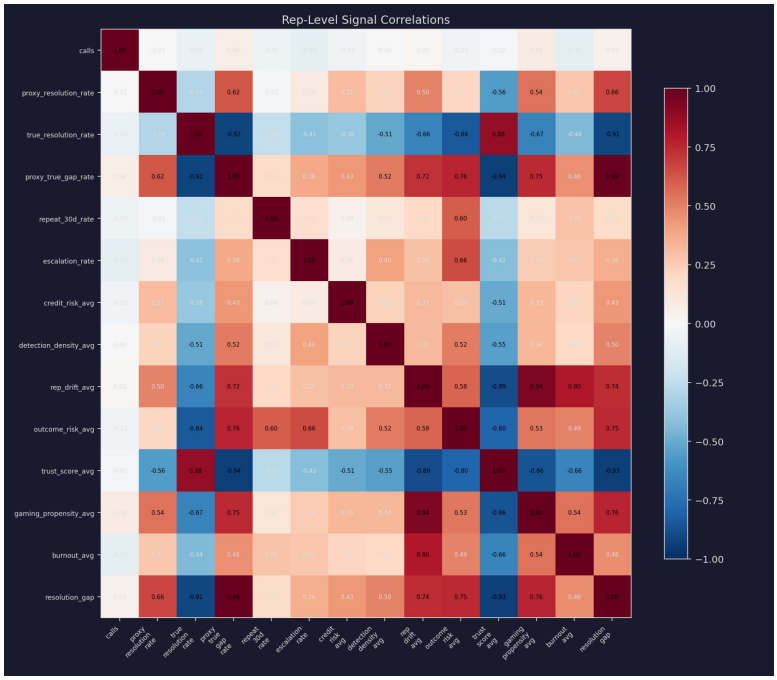

Figure 6. *Rep-level signal correlation matrix. Gaming propensity drives the resolution gap; call volume is independent of all trust signals.*

Monthly Trust Score Trend

Figure 7 tracks the mean trust score and drift velocity across 12 months. The trust score remains remarkably stable, fluctuating between 70.2 and 71.4 across the observation window. Drift velocity (bars) oscillates around zero with no sustained negative trend. This confirms the steady-state Goodhart interpretation: the system is not progressively degrading but has instead reached an equilibrium where the proxy metric and the true outcome have permanently diverged. The P10 trust score (shaded area) similarly holds steady around 40–41, indicating that the worst-performing decile is not worsening over time.

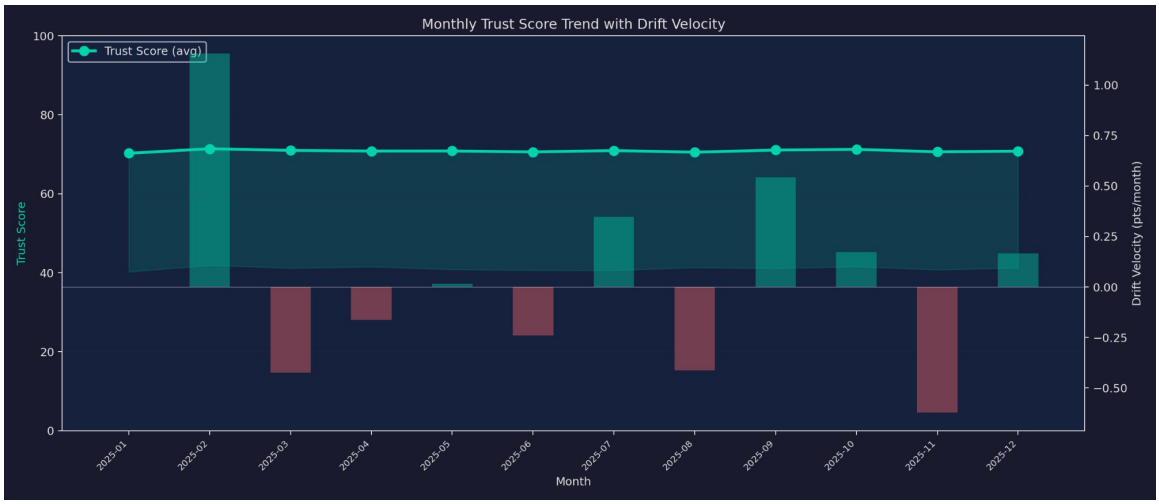

Figure 7. *Monthly trust score trend with drift velocity overlay. Flat trust trajectory and oscillating velocity confirm steady-state divergence.*

Monthly Governance Signal Trends

Figure 8 tracks four governance signals month-over-month: resolution gap (~42–44pp), churn rate (~26–27%), repeat contact rate (~19–20%), and bandaid rate (~8–9.5%). All four signals are flat across the 12-month window. The resolution gap does not widen or narrow; the bandaid rate does not escalate; the churn rate does not respond to proxy resolution improvements. This parallel stability is the signature of a system where the proxy metric has been fully decoupled from the underlying reality it was designed to measure.

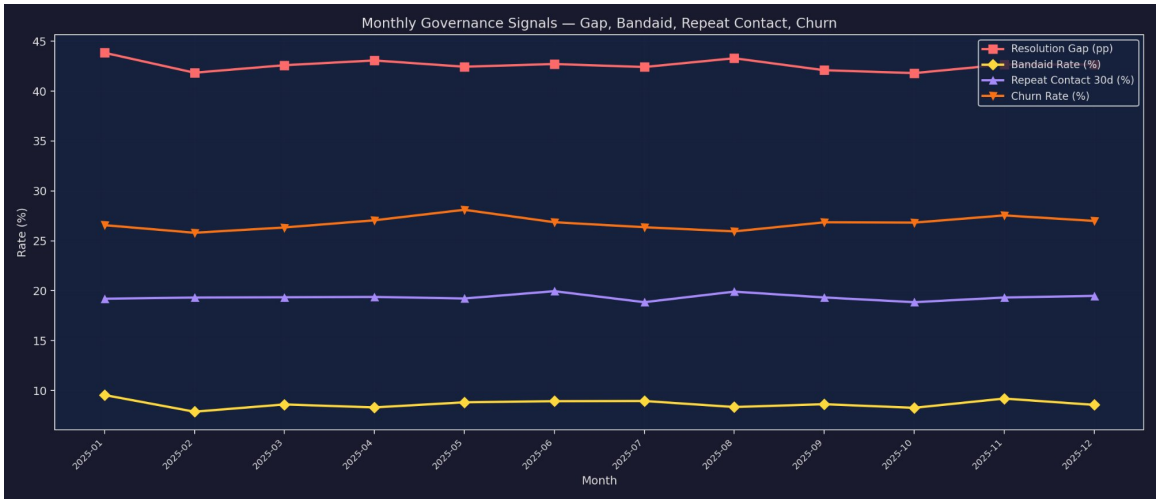

Figure 8. *Monthly governance signals: resolution gap, bandaid rate, repeat contacts, and churn rate. All four are stable, confirming metric decoupling.*

Churn Integration: Does Trust Predict Retention?

Figure 9 presents the critical churn integration test. Churn rate (bars) is plotted by trust score decile alongside the resolution gap (line). The trust-churn correlation at the decile level is $r = -0.328$—a weak negative relationship. Churn rates range from 25.7% (D6, trust 76–86) to 27.3% (D1, trust 18–41), a spread of only 1.6 percentage points across the full trust score range. Meanwhile, the resolution gap shows a dramatic cliff: deciles 1–4 carry gaps above 97pp (near-total proxy inflation), while deciles 7–10 show gaps near zero or slightly negative (true resolution exceeds proxy).

This result is consistent with the broader NovaWireless research finding that churn signal is structurally diffuse. The trust score successfully identifies calls where the proxy metric lies (the gap cliff is definitive), but this does not translate into churn prediction because churn is driven by account-level longitudinal factors—tenure, billing history, competitive offers—rather than call-level resolution outcomes. The trust score is *governmentally* useful (it tells you which labels to distrust) even though it is not *predictively* useful for churn.

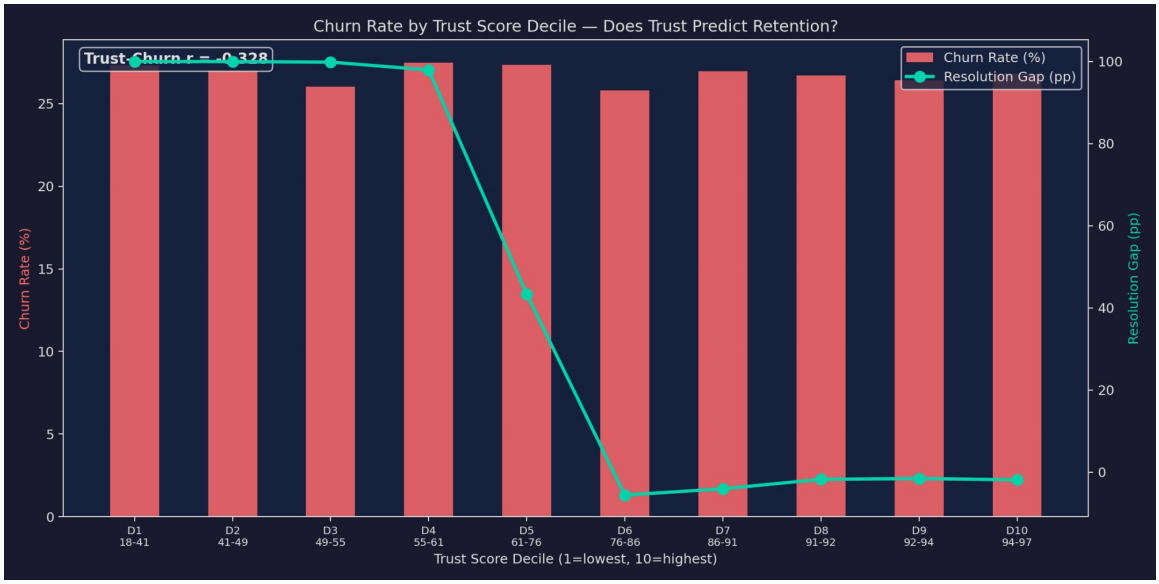

Figure 9. *Churn rate by trust score decile with resolution gap overlay. Trust-churn r = –0.328. The gap cliff is definitive; the churn signal is diffuse.*

Threshold Governance Framework

The pipeline evaluates configurable WATCH and VETO thresholds at three levels. At the scenario level, any scenario with trust score below 50 triggers VETO (halt AI optimization), below 65 triggers WATCH (human review), resolution gap above 70% triggers VETO, and bandaid rate above 50% triggers VETO. At the representative level, trust below 60 triggers VETO and below 65 triggers WATCH. Monthly drift velocity exceeding 2 points per month triggers WATCH.

Table 1

Governance Alert Summary for Current Dataset

Alert Level	Entity Type	Count	Example
VETO	Scenario	7	gamed_metric trust 46.0 < 50
VETO	Scenario		fraud_hic_exchange gap 80.7% > 70%
VETO	Scenario		gamed_metric bandaid 70.9% > 50%
WATCH	Scenario	3	activation_failed trust 64.8 < 65
WATCH	Rep	~50	REP00090 trust 64.9 < 65

Note. VETO alerts indicate that AI optimization labels from the affected entity should not be used. WATCH alerts require human review before proceeding.

Discussion

The pipeline confirms and extends the findings of the companion *When KPIs Lie* and *Goodhart's Law Addendum* papers through an independent analytical pathway. Where those papers demonstrated the gap's existence and stability, this pipeline provides the operational machinery to *act on* the gap in real time. The WATCH/VETO framework translates research findings into governance decisions: when the pipeline says VETO on gamed_metric, it means that any AI system trained on labels from that scenario is learning to replicate gaming behavior rather than genuine resolution.

The churn integration result ($r = -0.328$) deserves careful interpretation. A naive reading might conclude that the trust score "doesn't work" because it doesn't predict churn. The correct interpretation is that the trust score successfully measures what it was designed to measure—proxy-outcome divergence—and that this divergence is structurally unrelated to churn at the call level. Churn requires longitudinal, account-level signals. The trust score's value is governance, not prediction.

The monthly stability finding (Figures 7–8) has implications for AI governance more broadly. Systems that are progressively degrading are easier to detect because the trend creates urgency. Systems that reach a stable Goodhart equilibrium are harder to detect because nothing appears to be changing—the dashboard is green, the KPIs are stable, and the divergence hides in plain sight. The pipeline's monthly tracking capability is designed to detect both patterns: progressive degradation through drift velocity alerts, and stable divergence through absolute threshold alerts on the gap itself.

Ecosystem Position

This pipeline occupies a specific role in the NovaWireless research ecosystem. The *novawireless-lab* generates synthetic data upstream. This pipeline provides structured signal diagnostics through call metadata. The *novawireless-transcript-analysis* repository provides independent linguistic signal diagnostics through NLP features. The *NovaWireless_KPI_Drift_Observatory* produces the formal System Integrity Index and hardening extensions. Together, these four repositories form a complete chain from data generation through multi-modal governance to formal integrity assessment.

References

Aulabaugh, G. (2026a). When KPIs lie: A pressure experiment on proxy metric integrity in the NovaWireless synthetic call center. *NovaWireless Synthetic Research Lab.*

Aulabaugh, G. (2026b). Goodhart's Law addendum: 12-month KPI drift observation in the NovaWireless synthetic call center. *NovaWireless Synthetic Research Lab.*

Aulabaugh, G. (2026c). Trust signal health assessment of the NovaWireless synthetic call center: A diagnostic pipeline for detecting proxy-outcome divergence. PixelKraze LLC.

Goodhart, C. A. E. (1975). Problems of monetary management: The U.K. experience. In A. S. Courakis (Ed.), *Inflation, depression, and economic policy in the West* (pp. 116–132). Rowman & Littlefield.

Strathern, M. (1997). 'Improving ratings': Audit in the British university system. *European Review*, *5*(3), 305–321.